GLOBAL EDITION

Assembly Language

for x86 Processors

SEVENTH EDITION

Kip R. Irvine

ALWAYS LEARNING

PEARSON

Assembly Language for x86 Processors

Seventh Edition
Global Edition

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Florida International University
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Global Edition contributions by

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Essex CM20 2JE
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and Associated Companies throughout the world

Visit us on the World Wide Web at:
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Previously published as Assembly Language for Intel-Based Computers.

ISBN 10: 1-292-06121-9
ISBN 13: 978-1-292-06121-4 (Print)
ISBN 13: 978-1-292-06655-4 (PDF)

British Library Cataloguing-in-Publication Data
A catalogue record for this book is available from the British Library

10 9 8 7 6 5 4 3 2 1
14 13 12 11 10

Typeset in Times by Pavithra Jayapaul, Jouve
Printed and bound by Courier/Westford in The United States of America

To Jack and Candy Irvine

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Contents

Chapters are available from the Companion Web site

Appendices are available from the Companion Web site

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Preface

Assembly Language for x86 Processors, Seventh Edition, teaches assembly language programming and architecture for x86 and Intel64 processors. It is an appropriate text for the following types of college courses:

- Assembly Language Programming
- Fundamentals of Computer Systems
- Fundamentals of Computer Architecture

Students use Intel or AMD processors and program with **Microsoft Macro Assembler (MASM)**, running on recent versions of Microsoft Windows. Although this book was originally designed as a programming textbook for college students, it serves as an effective supplement to computer architecture courses. As a testament to its popularity, previous editions have been translated into numerous languages.

Emphasis of Topics This edition includes topics that lead naturally into subsequent courses in computer architecture, operating systems, and compiler writing:

- Virtual machine concept
- Instruction set architecture
- Elementary Boolean operations
- Instruction execution cycle
- Memory access and handshaking
- Interrupts and polling
- Hardware-based I/O
- Floating-point binary representation

Other topics relate specially to x86 and Intel64 architecture:

- Protected memory and paging
- Memory segmentation in real-address mode
- 16-Bit interrupt handling
- MS-DOS and BIOS system calls (interrupts)
- Floating-point unit architecture and programming
- Instruction encoding

Certain examples presented in the book lend themselves to courses that occur later in a computer science curriculum:

- Searching and sorting algorithms
- High-level language structures

- Finite-state machines
- Code optimization examples

What's New in the Seventh Edition

In this revision, we increased the discussions of program examples early in the book, added more supplemental review questions and key terms, introduced 64-bit programming, and reduced our dependence on the book's subroutine library. To be more specific, here are the details:

- Early chapters now include short sections that feature 64-bit CPU architecture and programming, and we have created a 64-bit version of the book's subroutine library named *Irvine64*.
- Many of the review questions and exercises have been modified, replaced, and moved from the middle of the chapter to the end of chapters, and divided into two sections: (1) Short answer questions, and (2) Algorithm workbench exercises. The latter exercises require the student to write a short amount of code to accomplish a goal.
- Each chapter now has a *Key Terms* section, listing new terms and concepts, as well as new MASM directives and Intel instructions.
- New programming exercises have been added, others removed, and a few existing exercises were modified.
- There is far less dependency on the author's subroutine libraries in this edition. Students are encouraged to call system functions themselves and use the Visual Studio debugger to step through the programs. The Irvine32 and Irvine64 libraries are available to help students handle input/output, but their use is not required.
- New tutorial videos covering essential content topics have been created by the author and added to the Pearson website.

This book is still focused on its primary goal, to teach students how to write and debug programs at the machine level. It will never replace a complete book on computer architecture, but it does give students the first-hand experience of writing software in an environment that teaches them how a computer works. Our premise is that students retain knowledge better when theory is combined with experience. In an engineering course, students construct prototypes; in a computer architecture course, students should write machine-level programs. In both cases, they have a memorable experience that gives them the confidence to work in any OS/machine-oriented environment.

Protected mode programming is entirely the focus of the printed chapters (1 through 13). As such, students will create 32-bit and 64-bit programs that run under the most recent versions of Microsoft Windows. The remaining four chapters cover 16-bit programming, and are supplied in electronic form. These chapters cover BIOS programming, MS-DOS services, keyboard and mouse input, video programming, and graphics. One chapter covers disk storage fundamentals. Another chapter covers advanced DOS programming techniques.

Subroutine Libraries We supply three versions of the subroutine library that students use for basic input/output, simulations, timing, and other useful tasks. The Irvine32 and Irvine64 libraries run in protected mode. The 16-bit version (Irvine16.lib) runs in real-address mode and is used only by Chapters 14 through 17. Full source code for the libraries is supplied on the companion website. The link libraries are available only for convenience, not to prevent students from learning how to program input–output themselves. Students are encouraged to create their own libraries.

Included Software and Examples All the example programs were tested with Microsoft Macro Assembler Version 11.0, running in Microsoft Visual Studio 2012. In addition, batch files are supplied that permit students to assemble and run applications from the Windows command

prompt. The 32-bit C++ applications in Chapter 14 were tested with Microsoft Visual C++ .NET. Information Updates and corrections to this book may be found at the Companion Web site, including additional programming projects for instructors to assign at the ends of chapters.

Overall Goals

The following goals of this book are designed to broaden the student's interest and knowledge in topics related to assembly language:

- Intel and AMD processor architecture and programming
- Real-address mode and protected mode programming
- Assembly language directives, macros, operators, and program structure
- Programming methodology, showing how to use assembly language to create system-level software tools and application programs
- Computer hardware manipulation
- Interaction between assembly language programs, the operating system, and other application programs

One of our goals is to help students approach programming problems with a machine-level mind set. It is important to think of the CPU as an interactive tool, and to learn to monitor its operation as directly as possible. A debugger is a programmer's best friend, not only for catching errors, but as an educational tool that teaches about the CPU and operating system. We encourage students to look beneath the surface of high-level languages and to realize that most programming languages are designed to be portable and, therefore, independent of their host machines. In addition to the short examples, this book contains hundreds of ready-to-run programs that demonstrate instructions or ideas as they are presented in the text. Reference materials, such as guides to MS-DOS interrupts and instruction mnemonics, are available at the end of the book.

Required Background The reader should already be able to program confidently in at least one high-level programming language such as Python, Java, C, or C++. One chapter covers C++ interfacing, so it is very helpful to have a compiler on hand. I have used this book in the classroom with majors in both computer science and management information systems, and it has been used elsewhere in engineering courses.

Features

Complete Program Listings The Companion Web site contains supplemental learning materials, study guides, and all the source code from the book's examples. An extensive link library is supplied with the book, containing more than 30 procedures that simplify user input–output, numeric processing, disk and file handling, and string handling. In the beginning stages of the course, students can use this library to enhance their programs. Later, they can create their own procedures and add them to the library.

Programming Logic Two chapters emphasize Boolean logic and bit-level manipulation. A conscious attempt is made to relate high-level programming logic to the low-level details of the machine. This approach helps students to create more efficient implementations and to better understand how compilers generate object code.

Hardware and Operating System Concepts The first two chapters introduce basic hardware and data representation concepts, including binary numbers, CPU architecture, status flags, and memory mapping. A survey of the computer's hardware and a historical perspective of the Intel processor family helps students to better understand their target computer system.

Structured Programming Approach Beginning with Chapter 5, procedures and functional decomposition are emphasized. Students are given more complex programming exercises, requiring them to focus on design before starting to write code.

Java Bytecodes and the Java Virtual Machine In Chapters 8 and 9, the author explains the basic operation of Java bytecodes with short illustrative examples. Numerous short examples are shown in disassembled bytecode format, followed by detailed step-by-step explanations.

Disk Storage Concepts Students learn the fundamental principles behind the disk storage system on MS-Windows–based systems from hardware and software points of view.

Creating Link Libraries Students are free to add their own procedures to the book's link library and create new libraries. They learn to use a toolbox approach to programming and to write code that is useful in more than one program.

Macros and Structures A chapter is devoted to creating structures, unions, and macros, which are essential in assembly language and systems programming. Conditional macros with advanced operators serve to make the macros more professional.

Interfacing to High-Level Languages A chapter is devoted to interfacing assembly language to C and C++. This is an important job skill for students who are likely to find jobs programming in high-level languages. They can learn to optimize their code and see examples of how C++ compilers optimize code.

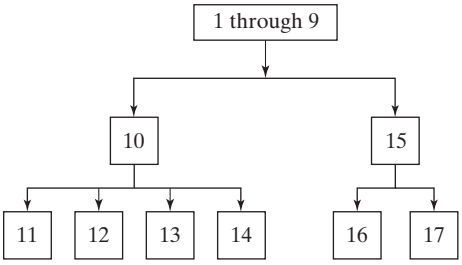
Instructional Aids All the program listings are available on the Web. Instructors are provided a test bank, answers to review questions, solutions to programming exercises, and a Microsoft PowerPoint slide presentation for each chapter.

VideoNotes VideoNotes are Pearson's new visual tool designed to teach students key programming concepts and techniques. These short step-by-step videos demonstrate basic assembly language concepts. VideoNotes allow for self-paced instruction with easy navigation including the ability to select, play, rewind, fast-forward, and stop within each VideoNote exercise.

VideoNotes are free with the purchase of a new textbook. To access VideoNotes, go to www.pearsonglobaleditions.com/irvine and click on the VideoNotes under *Student Resources*.

Chapter Descriptions

Chapters 1 to 8 contain core concepts of assembly language and should be covered in sequence. After that, you have a fair amount of freedom. The following chapter dependency graph shows how later chapters depend on knowledge gained from other chapters.



1. **Basic Concepts:** Applications of assembly language, basic concepts, machine language, and data representation.
2. **x86 Processor Architecture:** Basic microcomputer design, instruction execution cycle, x86 processor architecture, Intel64 architecture, x86 memory management, components of a microcomputer, and the input–output system.
3. **Assembly Language Fundamentals:** Introduction to assembly language, linking and debugging, and defining constants and variables.
4. **Data Transfers, Addressing, and Arithmetic:** Simple data transfer and arithmetic instructions, assemble-link-execute cycle, operators, directives, expressions, JMP and LOOP instructions, and indirect addressing.
5. **Procedures:** Linking to an external library, description of the book's link library, stack operations, defining and using procedures, flowcharts, and top-down structured design.
6. **Conditional Processing:** Boolean and comparison instructions, conditional jumps and loops, high-level logic structures, and finite-state machines.
7. **Integer Arithmetic:** Shift and rotate instructions with useful applications, multiplication and division, extended addition and subtraction, and ASCII and packed decimal arithmetic.
8. **Advanced Procedures:** Stack parameters, local variables, advanced PROC and INVOKE directives, and recursion.
9. **Strings and Arrays:** String primitives, manipulating arrays of characters and integers, two-dimensional arrays, sorting, and searching.
10. **Structures and Macros:** Structures, macros, conditional assembly directives, and defining repeat blocks.
11. **MS-Windows Programming:** Protected mode memory management concepts, using the Microsoft-Windows API to display text and colors, and dynamic memory allocation.
12. **Floating-Point Processing and Instruction Encoding:** Floating-point binary representation and floating-point arithmetic. Learning to program the IA-32 floating-point unit. Understanding the encoding of IA-32 machine instructions.
13. **High-Level Language Interface:** Parameter passing conventions, inline assembly code, and linking assembly language modules to C and C++ programs.

- **Appendix A:** MASM Reference
- **Appendix B:** The x86 Instruction Set
- **Appendix C:** Answers to Review Questions

The following chapters and appendices are supplied online at the Companion Web site:

14. **16-Bit MS-DOS Programming:** Memory organization, interrupts, function calls, and standard MS-DOS file I/O services.
15. **Disk Fundamentals:** Disk storage systems, sectors, clusters, directories, file allocation tables, handling MS-DOS error codes, and drive and directory manipulation.
16. **BIOS-Level Programming:** Keyboard input, video text, graphics, and mouse programming.
17. **Expert MS-DOS Programming:** Custom-designed segments, runtime program structure, and Interrupt handling. Hardware control using I/O ports.

- **Appendix D:** BIOS and MS-DOS Interrupts
- **Appendix E:** Answers to Review Questions (Chapters 14–17)

Instructor and Student Resources

Instructor Resource Materials

The following protected instructor material is available on the Companion Web site:

www.pearsonglobaleditions.com/irvine

For username and password information, please contact your Pearson Representative.

- Lecture PowerPoint Slides
- Instructor Solutions Manual

Student Resource Materials

The student resource materials can be accessed through the publisher's Web site located at *www.pearsonglobaleditions.com/irvine*. These resources include:

- VideoNotes
- Online Chapters and Appendices
  - Chapter 14: *16-Bit MS-DOS Programming*
  - Chapter 15: *Disk Fundamentals*
  - Chapter 16: *BIOS-Level Programming*
  - Chapter 17: *Expert MS-DOS Programming*
  - Appendix D: *BIOS and MS-DOS Interrupts*
  - Appendix E: *Answers to Review Questions (Chapters 14–17)*

Students must use the access card located in the front of the book to register and access the online chapters and VideoNotes. Instructors must also register on the site to access this material. Students will also find a link to the author's Web site. An access card is not required for the following materials, located at *www.asmirvine.com*:

- *Getting Started*, a comprehensive step-by-step tutorial that helps students customize Visual Studio for assembly language programming.
- Supplementary articles on assembly language programming topics.
- Complete source code for all example programs in the book, as well as the source code for the author's supplementary library.

- *Assembly Language Workbook*, an interactive workbook covering number conversions, addressing modes, register usage, debug programming, and floating-point binary numbers. Content pages are HTML documents to allow for customization. Help File in Windows Help Format.
- Debugging Tools: Tutorials on using the Microsoft Visual Studio debugger.

Acknowledgments

Many thanks are due to Tracy Johnson, Executive Editor for Computer Science at Pearson Education, who has provided friendly, helpful guidance over the past few years. Pavithra Jayapaul of Jouve did an excellent job on the book production, along with Greg Dulles as the production editor at Pearson.

Previous Editions

I offer my special thanks to the following individuals who were most helpful during the development of earlier editions of this book:

- William Barrett, San Jose State University
- Scott Blackledge
- James Brink, Pacific Lutheran University
- Gerald Cahill, Antelope Valley College
- John Taylor

Global Edition

Pearson would like to thank and acknowledge the following people for reviewing the Global Edition:

- Agnimitra Borah, Girijananda Chowdhury Institute of Management and Technology
- Supratim Gupta, Indian Institute of Technology Kharagpur
- Chiranjib Koley, National Institute of Technology Durgapur

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About the Author

Kip Irvine has written five computer programming textbooks, for Intel Assembly Language, C++, Visual Basic (beginning and advanced), and COBOL. His book *Assembly Language for Intel-Based Computers* has been translated into six languages. His first college degrees (B.M., M.M., and doctorate) were in Music Composition, at University of Hawaii and University of Miami. He began programming computers for music synthesis around 1982 and taught programming at Miami-Dade Community College for 17 years. Kip earned an M.S. degree in Computer Science from the University of Miami, and he has been a full-time member of the faculty in the School of Computing and Information Sciences at Florida International University since 2000.

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1

Basic Concepts

This chapter establishes some core concepts relating to assembly language programming. For example, it shows how assembly language fits into the wide spectrum of languages and applications. We introduce the virtual machine concept, which is so important in understanding the relationship between software and hardware layers. A large part of the chapter is devoted to the binary and hexadecimal numbering systems, showing how to perform conversions and do basic arithmetic. Finally, this chapter introduces fundamental boolean operations (AND, OR, NOT, XOR), which will prove to be essential in later chapters.

1.1 Welcome to Assembly Language

Assembly Language for x86 Processors focuses on programming microprocessors compatible with Intel and AMD processors running under 32-bit and 64-bit versions of Microsoft Windows.

The latest version of *Microsoft Macro Assembler* (known as *MASM*) should be used with this book. MASM is included with most versions of Microsoft Visual Studio (Pro, Ultimate, Express, . . .). Please check our web site (*asmirvine.com*) for the latest details about support for MASM in Visual Studio. We also include lots of helpful information about how to set up your software and get started.

Some other well-known assemblers for x86 systems running under Microsoft Windows include TASM (Turbo Assembler), NASM (Netwide Assembler), and MASM32 (a variant of MASM). Two popular Linux-based assemblers are GAS (GNU assembler) and NASM. Of these, NASM's syntax is most similar to that of MASM.

Assembly language is the oldest programming language, and of all languages, bears the closest resemblance to native machine language. It provides direct access to computer hardware, requiring you to understand much about your computer's architecture and operating system.

Educational Value Why read this book? Perhaps you're taking a college course whose title is similar to one of the following courses that often use our book:

- Microcomputer Assembly Language
- Assembly Language Programming
- Introduction to Computer Architecture
- Fundamentals of Computer Systems
- Embedded Systems Programming

This book will help you learn basic principles about computer architecture, machine language, and low-level programming. You will learn enough assembly language to test your knowledge on today's most widely used microprocessor family. You won't be learning to program a "toy" computer using a simulated assembler; MASM is an industrial-strength assembler, used by practicing professionals. You will learn the architecture of the Intel processor family from a programmer's point of view.

If you are planning to be a C or C++ developer, you need to develop an understanding of how memory, address, and instructions work at a low level. A lot of programming errors are not easily recognized at the high-level language level. You will often find it necessary to "drill down" into your program's internals to find out why it isn't working.

If you doubt the value of low-level programming and studying details of computer software and hardware, take note of the following quote from a leading computer scientist, Donald Knuth, in discussing his famous book series, *The Art of Computer Programming*:

> Some people [say] that having machine language, at all, was the great mistake that I made. I really don't think you can write a book for serious computer programmers unless you are able to discuss low-level detail.[1]

Visit this book's web site to get lots of supplemental information, tutorials, and exercises at **www.asmirvine.com**

1.1.1 Questions You Might Ask

What Background Should I Have? Before reading this book, you should have programmed in at least one structured high-level language, such as Java, C, Python, or C++. You should know how to use IF statements, arrays, and functions to solve programming problems.

What Are Assemblers and Linkers? An *assembler* is a utility program that converts source code programs from assembly language into machine language. A *linker* is a utility program that combines individual files created by an assembler into a single executable program. A related utility, called a *debugger*, lets you to step through a program while it's running and examine registers and memory.

What Hardware and Software Do I Need? You need a computer that runs a 32-bit or 64-bit version of Microsoft Windows, along with one of the recent versions of Microsoft Visual Studio.

What Types of Programs Can Be Created Using MASM?

- *32-Bit Protected Mode:* 32-bit protected mode programs run under all 32-bit versions of Microsoft Windows. They are usually easier to write and understand than real-mode programs. From now on, we will simply call this *32-bit mode*.
- *64-Bit Mode:* 64-bit programs run under all 64-bit versions of Microsoft Windows.
- *16-Bit Real-Address Mode:* 16-bit programs run under 32-bit versions of Windows and on embedded systems. Because they are not supported by 64-bit Windows, we will restrict discussions of this mode to Chapters 14 through 17. These chapters are in electronic form, available from the publisher's web site.

What Supplements Are Supplied with This Book? The book's web site (*www.asmirvine.com*) has the following:

- ***Assembly Language Workbook***, a collection of tutorials
- ***Irvine32, Irvine64, and Irvine16 subroutine libraries*** for 64-bit, 32-bit, and 16-bit programming, with complete source code
- ***Example programs*** with all source code from the book
- ***Corrections*** to the book
- ***Getting Started***, a detailed tutorial designed to help you set up Visual Studio to use the Microsoft assembler
- ***Articles*** on advanced topics not included in the printed book for lack of space
- ***A link to an online discussion forum***, where you can get help from other experts who use the book

What Will I Learn? This book should make you better informed about data representation, debugging, programming, and hardware manipulation. Here's what you will learn:

- Basic principles of computer architecture as applied to x86 processors
- Basic boolean logic and how it applies to programming and computer hardware
- How x86 processors manage memory, using protected mode and virtual mode
- How high-level language compilers (such as C++) translate statements from their language into assembly language and native machine code

- How high-level languages implement arithmetic expressions, loops, and logical structures at the machine level
- Data representation, including signed and unsigned integers, real numbers, and character data
- How to debug programs at the machine level. The need for this skill is vital when you work in languages such as C and C++, which generate native machine code
- How application programs communicate with the computer's operating system via interrupt handlers and system calls
- How to interface assembly language code to C++ programs
- How to create assembly language application programs

How Does Assembly Language Relate to Machine Language? *Machine language* is a numeric language specifically understood by a computer's processor (the CPU). All x86 processors understand a common machine language. *Assembly language* consists of statements written with short mnemonics such as ADD, MOV, SUB, and CALL. Assembly language has a *one-to-one* relationship with machine language: Each assembly language instruction corresponds to a single machine-language instruction.

How Do C++ and Java Relate to Assembly Language? High-level languages such as Python, C++, and Java have a *one-to-many* relationship with assembly language and machine language. A single statement in C++, for example, expands into multiple assembly language or machine instructions. Most people cannot read raw machine code, so in this book, we examine its closest relative, assembly language. For example, the following C++ code carries out two arithmetic operations and assigns the result to a variable. Assume X and Y are integers:

```
int   Y;
int   X = (Y + 4) * 3;
```

Following is the equivalent translation to assembly language. The translation requires multiple statements because each assembly language statement corresponds to a single machine instruction:

```
mov   eax,Y                    ; move Y to the EAX register
add   eax,4                    ; add 4 to the EAX register
mov   ebx,3                    ; move 3 to the EBX register
imul  ebx                      ; multiply EAX by EBX
mov   X,eax                    ; move EAX to X
```

(*Registers* are named storage locations in the CPU that hold intermediate results of operations.) The point of this example is not to claim that C++ is superior to assembly language or vice versa, but to show their relationship.

Is Assembly Language Portable? A language whose source programs can be compiled and run on a wide variety of computer systems is said to be *portable*. A C++ program, for example, will compile and run on just about any computer, unless it makes specific references to library functions that exist under a single operating system. A major feature of the Java language is that compiled programs run on nearly any computer system.

Assembly language is not portable, because it is designed for a specific processor family. There are a number of different assembly languages widely used today, each based on a processor family.

Some well-known processor families are Motorola 68x00, x86, SUN Sparc, Vax, and IBM-370. The instructions in assembly language may directly match the computer's architecture or they may be translated during execution by a program inside the processor known as a *microcode interpreter*.

Why Learn Assembly Language? If you're still not convinced that you should learn assembly language, consider the following points:

- If you study computer engineering, you may likely be asked to write *embedded* programs. They are short programs stored in a small amount of memory in single-purpose devices such as telephones, automobile fuel and ignition systems, air-conditioning control systems, security systems, data acquisition instruments, video cards, sound cards, hard drives, modems, and printers. Assembly language is an ideal tool for writing embedded programs because of its economical use of memory.
- Real-time applications dealing with simulation and hardware monitoring require precise timing and responses. High-level languages do not give programmers exact control over machine code generated by compilers. Assembly language permits you to precisely specify a program's executable code.
- Computer game consoles require their software to be highly optimized for small code size and fast execution. Game programmers are experts at writing code that takes full advantage of hardware features in a target system. They often use assembly language as their tool of choice because it permits direct access to computer hardware, and code can be hand optimized for speed.
- Assembly language helps you to gain an overall understanding of the interaction between computer hardware, operating systems, and application programs. Using assembly language, you can apply and test theoretical information you are given in computer architecture and operating systems courses.
- Some high-level languages abstract their data representation to the point that it becomes awkward to perform low-level tasks such as bit manipulation. In such an environment, programmers will often call subroutines written in assembly language to accomplish their goal.
- Hardware manufacturers create device drivers for the equipment they sell. *Device drivers* are programs that translate general operating system commands into specific references to hardware details. Printer manufacturers, for example, create a different MS-Windows device driver for each model they sell. Often these device drivers contain significant amounts of assembly language code.

Are There Rules in Assembly Language? Most rules in assembly language are based on physical limitations of the target processor and its machine language. The CPU, for example, requires two instruction operands to be the same size. Assembly language has fewer rules than C++ or Java because the latter use syntax rules to reduce unintended logic errors at the expense of low-level data access. Assembly language programmers can easily bypass restrictions characteristic of high-level languages. Java, for example, does not permit access to specific memory addresses. One can work around the restriction by calling a C function using JNI (*Java Native Interface*) classes, but the resulting program can be awkward to maintain. Assembly language, on the other hand, can access any memory address. The price for such freedom is high: Assembly language programmers spend a lot of time debugging!

1.1.2 Assembly Language Applications

In the early days of programming, most applications were written partially or entirely in assembly language. They had to fit in a small area of memory and run as efficiently as possible on slow processors. As memory became more plentiful and processors dramatically increased in speed, programs became more complex. Programmers switched to high-level languages such as C, FORTRAN, and COBOL that contained a certain amount of structuring capability. More recently, object-oriented languages such as Python, C++, C#, and Java have made it possible to write complex programs containing millions of lines of code.

It is rare to see large application programs coded completely in assembly language because they would take too much time to write and maintain. Instead, assembly language is used to optimize certain sections of application programs for speed and to access computer hardware. Table 1-1 compares the adaptability of assembly language to high-level languages in relation to various types of applications.

Table 1-1 Comparison of Assembly Language to High-Level Languages.

Type of Application	High-Level Languages	Assembly Language
Commercial or scientific application, written for single platform, medium to large size.	Formal structures make it easy to organize and maintain large sections of code.	Minimal formal structure, so one must be imposed by programmers who have varying levels of experience. This leads to difficulties maintaining existing code.
Hardware device driver.	The language may not provide for direct hardware access. Even if it does, awkward coding techniques may be required, resulting in maintenance difficulties.	Hardware access is straightforward and simple. Easy to maintain when programs are short and well documented.
Commercial or scientific application written for multiple platforms (different operating systems).	Usually portable. The source code can be recompiled on each target operating system with minimal changes.	Must be recoded separately for each platform, using an assembler with a different syntax. Difficult to maintain.
Embedded systems and computer games requiring direct hardware access.	May produce large executable files that exceed the memory capacity of the device.	Ideal, because the executable code is small and runs quickly.

The C and C++ languages have the unique quality of offering a compromise between high-level structure and low-level details. Direct hardware access is possible but completely nonportable. Most C and C++ compilers allow you to embed assembly language statements in their code, providing access to hardware details.

1.1.3 Section Review

1. How do assemblers and linkers work together?
2. How will studying assembly language enhance your understanding of operating systems?

3. What is meant by a *one-to-many relationship* when comparing a high-level language to machine language?
4. Explain the concept of *portability* as it applies to programming languages.
5. Is the assembly language for x86 processors the same as those for computer systems such as the Vax or Motorola 68x00?
6. Give an example of an embedded systems application.
7. What is a device driver?
8. Do you suppose type checking on pointer variables is stronger (stricter) in assembly language, or in C and C++?
9. Name two types of applications that would be better suited to assembly language than a high-level language.
10. Why would a high-level language not be an ideal tool for writing a program that directly accesses a printer port?
11. Why is assembly language not usually used when writing large application programs?
12. *Challenge:* Translate the following C++ expression to assembly language, using the example presented earlier in this chapter as a guide: X = (Y * 4) + 3.

1.2 Virtual Machine Concept

An effective way to explain how a computer's hardware and software are related is called the *virtual machine concept*. A well-known explanation of this model can be found in Andrew Tanenbaum's book, *Structured Computer Organization.* To explain this concept, let us begin with the most basic function of a computer, executing programs.

A computer can usually execute programs written in its native *machine language*. Each instruction in this language is simple enough to be executed using a relatively small number of electronic circuits. For simplicity, we will call this language **L0**.

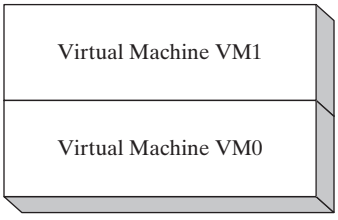
Programmers would have a difficult time writing programs in L0 because it is enormously detailed and consists purely of numbers. If a new language, **L1**, could be constructed that was easier to use, programs could be written in L1. There are two ways to achieve this:

- *Interpretation:* As the L1 program is running, each of its instructions could be decoded and executed by a program written in language L0. The L1 program begins running immediately, but each instruction has to be decoded before it can execute.
- *Translation:* The entire L1 program could be converted into an L0 program by an L0 program specifically designed for this purpose. Then the resulting L0 program could be executed directly on the computer hardware.

Virtual Machines

Rather than using only languages, it is easier to think in terms of a hypothetical computer, or *virtual machine*, at each level. Informally, we can define a virtual machine as a software program that emulates the functions of some other physical or virtual computer. The virtual machine

VM1, as we will call it, can execute commands written in language L1. The virtual machine **VM0** can execute commands written in language L0:



Each virtual machine can be constructed of either hardware or software. People can write programs for virtual machine VM1, and if it is practical to implement VM1 as an actual computer, programs can be executed directly on the hardware. Or programs written in VM1 can be interpreted/translated and executed on machine VM0.

Machine VM1 cannot be radically different from VM0 because the translation or interpretation would be too time-consuming. What if the language VM1 supports is still not programmer-friendly enough to be used for useful applications? Then another virtual machine, VM2, can be designed that is more easily understood. This process can be repeated until a virtual machine VM*n* can be designed to support a powerful, easy-to-use language.

The Java programming language is based on the virtual machine concept. A program written in the Java language is translated by a Java compiler into *Java byte code*. The latter is a low-level language quickly executed at runtime by a program known as a *Java virtual machine (JVM)*. The JVM has been implemented on many different computer systems, making Java programs relatively system independent.

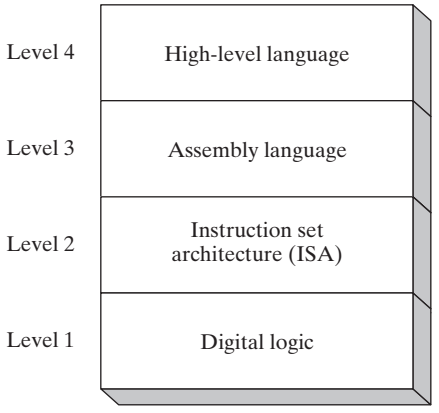
Specific Machines

Let us relate this to actual computers and languages, using names such as **Level 2** for VM2 and **Level 1** for VM1, shown in Figure 1-1. A computer's digital logic hardware represents machine Level 1. Above this is Level 2, called the *instruction set Architecture (ISA)*. This is the first level at which users can typically write programs, although the programs consist of binary values called *machine language*.

Instruction Set Architecture (Level 2) Computer chip manufacturers design into the processor an instruction set to carry out basic operations, such as move, add, or multiply. This set of instructions is also referred to as *machine language*. Each machine-language instruction is executed either directly by the computer's hardware or by a program embedded in the microprocessor chip called a *microprogram*. A discussion of microprograms is beyond the scope of this book, but you can refer to Tanenbaum for more details.

Assembly Language (Level 3) Above the ISA level, programming languages provide translation layers to make large-scale software development practical. Assembly language, which appears at Level 3, uses short mnemonics such as ADD, SUB, and MOV, which are easily translated to the ISA level. Assembly language programs are translated (assembled) in their entirety into machine language before they begin to execute.

Figure 1–1 Virtual machine levels.



High-Level Languages (Level 4) At Level 4 are high-level programming languages such as C, C++, and Java. Programs in these languages contain powerful statements that translate into multiple assembly language instructions. You can see such a translation, for example, by examining the listing file output created by a C++ compiler. The assembly language code is automatically assembled by the compiler into machine language.

1.2.1 Section Review

1. In your own words, describe the *virtual machine* concept.
2. Why do you suppose translated programs often execute more quickly than interpreted ones?
3. *(True/False):* When an interpreted program written in language L1 runs, each of its instructions is decoded and executed by a program written in language L0.
4. Explain the importance of translation when dealing with languages at different virtual machine levels.
5. At which level does assembly language appear in the virtual machine example shown in this section?
6. What software utility permits compiled Java programs to run on almost any computer?
7. Name the four virtual machine levels named in this section, from lowest to highest.
8. Why don't programmers write applications in machine language?
9. Machine language is used at which level of the virtual machine shown in Figure 1-1?
10. Statements at the assembly language level of a virtual machine are translated into statements at which other level?

1.3 Data Representation

Assembly language programmers deal with data at the physical level, so they must be adept at examining memory and registers. Often, binary numbers are used to describe the contents of computer memory; at other times, decimal and hexadecimal numbers are used. You must develop

a certain fluency with number formats, so you can quickly translate numbers from one format to another.

Each numbering format, or system, has a *base*, or maximum number of symbols that can be assigned to a single digit. Table 1-2 shows the possible digits for the numbering systems used most commonly in hardware and software manuals. In the last row of the table, hexadecimal numbers use the digits 0 through 9 and continue with the letters A through F to represent decimal values 10 through 15. It is quite common to use hexadecimal numbers when showing the contents of computer memory and machine-level instructions.

1.3.1 Binary Integers

A computer stores instructions and data in memory as collections of electronic charges. Representing these entities with numbers requires a system geared to the concepts of *on* and *off* or *true* and *false*. *Binary numbers* are base 2 numbers, in which each binary digit (called a *bit*) is either 0 or 1. ***Bits*** are numbered sequentially starting at zero on the right side and increasing toward the left. The bit on the left is called the *most significant bit* (MSB), and the bit on the right is the *least significant bit* (LSB). The MSB and LSB bit numbers of a 16-bit binary number are shown in the following figure:

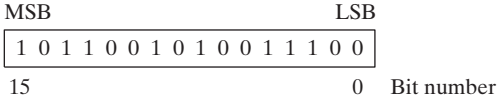


Table 1-2 Binary, Octal, Decimal, and Hexadecimal Digits.

System	Base	Possible Digits
Binary	2	0 1
Octal	8	0 1 2 3 4 5 6 7
Decimal	10	0 1 2 3 4 5 6 7 8 9
Hexadecimal	16	0 1 2 3 4 5 6 7 8 9 A B C D E F

Binary integers can be signed or unsigned. A signed integer is positive or negative. An unsigned integer is by default positive. Zero is considered positive. When writing down large binary numbers, many people like to insert a dot every 4 bits or 8 bits to make the numbers easier to read. Examples are 1101.1110.0011.1000.0000 and 11001010.10101100.

Unsigned Binary Integers

Starting with the LSB, each bit in an unsigned binary integer represents an increasing power of 2. The following figure contains an 8-bit binary number, showing how powers of two increase from right to left:

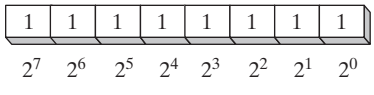


Table 1-3 lists the decimal values of 2^0 through 2^{15}.

Table 1-3 Binary Bit Position Values.

2^n	Decimal Value	2^n	Decimal Value
2^0	1	2^8	256
2^1	2	2^9	512
2^2	4	2^{10}	1024
2^3	8	2^{11}	2048
2^4	16	2^{12}	4096
2^5	32	2^{13}	8192
2^6	64	2^{14}	16384
2^7	128	2^{15}	32768

Translating Unsigned Binary Integers to Decimal

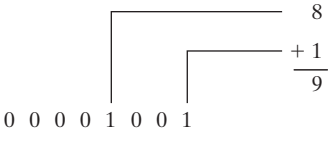
Weighted positional notation represents a convenient way to calculate the decimal value of an unsigned binary integer having n digits:

$$\text{dec} = (D_{n-1} \times 2^{n-1}) + (D_{n-2} \times 2^{n-2}) + \cdots + (D_1 \times 2^1) + (D_0 \times 2^0)$$

D indicates a binary digit. For example, binary 00001001 is equal to 9. We calculate this value by leaving out terms equal to zero:

$$(1 \times 2^3) + (1 \times 2^0) = 9$$

The same calculation is shown by the following figure:



Translating Unsigned Decimal Integers to Binary

To translate an unsigned decimal integer into binary, repeatedly divide the integer by 2, saving each remainder as a binary digit. The following table shows the steps required to translate decimal 37 to binary. The remainder digits, starting from the top row, are the binary digits D_0, D_1, D_2, D_3, D_4, and D_5:

Division	Quotient	Remainder
37 / 2	18	1
18 / 2	9	0
9 / 2	4	1
4 / 2	2	0
2 / 2	1	0
1 / 2	0	1

We can concatenate the binary bits from the remainder column of the table in reverse order (D_5, D_4, . . .) to produce binary 100101. Because computer storage always consists of binary numbers whose lengths are multiples of 8, we fill the remaining two digit positions on the left with zeros, producing 00100101.

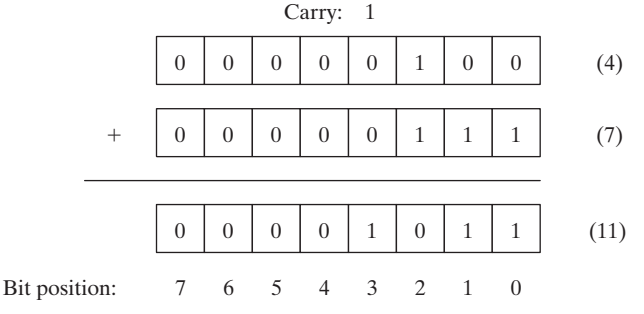
> ***Tip:*** How many bits? There's a simple formula to find b, the number of binary bits you need to represent the unsigned decimal value n. It is b = ceiling ($\log_2 n$). If n = 17, for example, $\log_2 17$ = 4.087463, which when raised to the smallest following integer, equals 5. Most calculators don't have a log base 2 operation, but you can find web pages that will calculate it for you.

1.3.2 Binary Addition

When adding two binary integers, proceed bit by bit, starting with the low-order pair of bits (on the right) and add each subsequent pair of bits. There are four ways to add two binary digits, as shown here:

0 + 0 = 0	0 + 1 = 1
1 + 0 = 1	1 + 1 = 10

When adding 1 to 1, the result is 10 binary (think of it as the decimal value 2). The extra digit generates a carry to the next-highest bit position. In the following figure, we add binary 00000100 to 00000111:



Beginning with the lowest bit in each number (bit position 0), we add 0 + 1, producing a 1 in the bottom row. The same happens in the next highest bit (position 1). In bit position 2, we add 1 + 1, generating a sum of zero and a carry of 1. In bit position 3, we add the carry bit to 0 + 0, producing 1. The rest of the bits are zeros. You can verify the addition by adding the decimal equivalents shown on the right side of the figure (4 + 7 = 11).

Sometimes a carry is generated out of the highest bit position. When that happens, the size of the storage area set aside becomes important. If we add 11111111 to 00000001, for example, a 1 carries out of the highest bit position, and the lowest 8 bits of the sum equal all zeros. If the storage location for the sum is at least 9 bits long, we can represent the sum as 100000000. But if the sum can only store 8 bits, it will equal to 00000000, the lowest 8 bits of the calculated value.

1.3.3 Integer Storage Sizes

The basic storage unit for all data in an x86 computer is a *byte*, containing 8 bits. Other storage sizes are *word* (2 bytes), *doubleword* (4 bytes), and *quadword* (8 bytes). In the following figure, the number of bits is shown for each size:

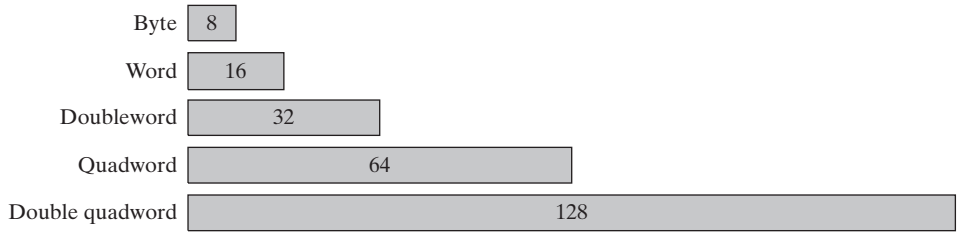


Table 1-4 shows the range of possible values for each type of unsigned integer.

Large Measurements A number of large measurements are used when referring to both memory and disk space:

- One *kilobyte* is equal to 2^{10}, or 1024 bytes.
- One *megabyte* (1 MByte) is equal to 2^{20}, or 1,048,576 bytes.
- One *gigabyte* (1 GByte) is equal to 2^{30}, or 1024^3, or 1,073,741,824 bytes.
- One *terabyte* (1 TByte) is equal to 2^{40}, or 1024^4, or 1,099,511,627,776 bytes.
- One *petabyte* is equal to 2^{50}, or 1,125,899,906,842,624 bytes.
- One *exabyte* is equal to 2^{60}, or 1,152,921,504,606,846,976 bytes.
- One *zettabyte* is equal to 2^{70} bytes.
- One *yottabyte* is equal to 2^{80} bytes.

Table 1-4 Ranges and Sizes of Unsigned Integer Types.

Type	Range	Storage Size in Bits
Unsigned byte	0 to $2^8 - 1$	8
Unsigned word	0 to $2^{16} - 1$	16
Unsigned doubleword	0 to $2^{32} - 1$	32
Unsigned quadword	0 to $2^{64} - 1$	64
Unsigned double quadword	0 to $2^{128} - 1$	128

1.3.4 Hexadecimal Integers

Large binary numbers are cumbersome to read, so hexadecimal digits offer a convenient way to represent binary data. Each digit in a hexadecimal integer represents four binary bits, and two hexadecimal digits together represent a byte. A single hexadecimal digit represents decimal 0 to 15, so letters A to F represent decimal values in the range 10 through 15. Table 1-5 shows how each sequence of four binary bits translates into a decimal or hexadecimal value.

Table 1-5 Binary, Decimal, and Hexadecimal Equivalents.

Binary	Decimal	Hexadecimal	Binary	Decimal	Hexadecimal
0000	0	0	1000	8	8
0001	1	1	1001	9	9
0010	2	2	1010	10	A
0011	3	3	1011	11	B
0100	4	4	1100	12	C
0101	5	5	1101	13	D
0110	6	6	1110	14	E
0111	7	7	1111	15	F

The following example shows how binary 0001 0110 1010 0111 1001 0100 is equivalent to hexadecimal 16A794:

1	6	A	7	9	4
0001	0110	1010	0111	1001	0100

Converting Unsigned Hexadecimal to Decimal

In hexadecimal, each digit position represents a power of 16. This is helpful when calculating the decimal value of a hexadecimal integer. Suppose we number the digits in a four-digit hexadecimal integer with subscripts as $D_3D_2D_1D_0$. The following formula calculates the integer's decimal value:

$$dec = (D_3 \times 16^3) + (D_2 \times 16^2) + (D_1 \times 16^1) + (D_0 \times 16^0)$$

The formula can be generalized for any *n*-digit hexadecimal integer:

$$dec = (D_{n-1} \times 16^{n-1}) + (D_{n-2} \times 16^{n-2}) + \cdots + (D_1 \times 16^1) + (D_0 \times 16^0)$$

> In general, you can convert an *n*-digit integer in any base B to decimal using the following formula: $dec = (D_{n-1} \times B^{n-1}) + (D_{n-2} \times B^{n-2}) + \cdots + (D_1 \times B^1) + (D_0 \times B^0)$.

For example, hexadecimal 1234 is equal to $(1 \times 16^3) + (2 \times 16^2) + (3 \times 16^1) + (4 \times 16^0)$, or decimal 4660. Similarly, hexadecimal 3BA4 is equal to $(3 \times 16^3) + (11 \times 16^2) + (10 \times 16^1) + (4 \times 16^0)$, or decimal 15,268. The following figure shows this last calculation:

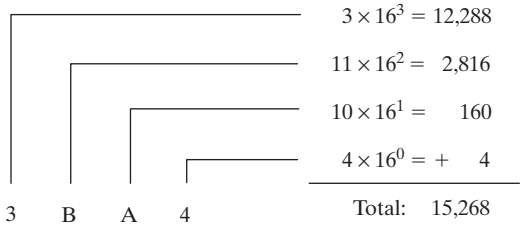


Table 1-6 lists the powers of 16 from 16^0 to 16^7.

Table 1-6 Powers of 16 in Decimal.

16^n	Decimal Value	16^n	Decimal Value
16^0	1	16^4	65,536
16^1	16	16^5	1,048,576
16^2	256	16^6	16,777,216
16^3	4096	16^7	268,435,456

Converting Unsigned Decimal to Hexadecimal

To convert an unsigned decimal integer to hexadecimal, repeatedly divide the decimal value by 16 and retain each remainder as a hexadecimal digit. For example, the following table lists the steps when converting decimal 422 to hexadecimal:

Division	Quotient	Remainder
422 / 16	26	6
26 / 16	1	A
1 / 16	0	1

The resulting hexadecimal number is assembled from the digits in the remainder column, starting from the last row and working upward to the top row. In this example, the hexadecimal representation is **1A6**. The same algorithm was used for binary integers in Section 1.3.1. To convert from decimal into some other number base other than hexadecimal, replace the divisor (16) in each calculation with the desired number base.

1.3.5 Hexadecimal Addition

Debugging utility programs (known as *debuggers*) usually display memory addresses in hexadecimal. It is often necessary to add two addresses in order to locate a new address. Fortunately, hexadecimal addition works the same way as decimal addition, if you just change the number base.

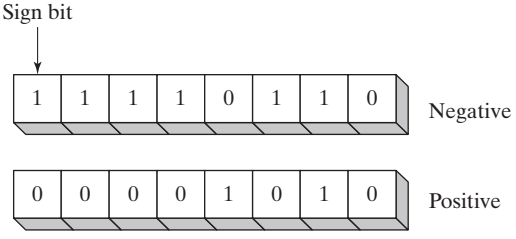
Suppose we want to add two numbers X and Y, using numbering base b. We will number their digits from the lowest position (x_0) to the highest. If we add digits x_i and y_i in X and Y, we produce the value s_i. If $s_i \geq b$, we recalculate $s_i = (s_i \text{ MOD } b)$ and generate a carry value of 1. When we move to the next pair of digits x_{i+1} and y_{i+1}, we add the carry value to their sum.

For example, let's add the hexadecimal values 6A2 and 49A. In the lowest digit position, 2 + A = decimal 12, so there is no carry and we use C to indicate the hexadecimal sum digit. In the next position, A + 9 = decimal 19, so there is a carry because $19 \geq 16$, the number base. We calculate 19 MOD 16 = 3, and carry a 1 into the third digit position. Finally, we add 1 + 6 + 4 = decimal 11, which is shown as the letter B in the third position of the sum. The hexadecimal sum is B3C.

Carry	1		
X	6	A	2
Y	4	9	A
S	B	3	C

1.3.6 Signed Binary Integers

Signed binary integers are positive or negative. For x86 processors, the MSB indicates the sign: 0 is positive and 1 is negative. The following figure shows examples of 8-bit negative and positive integers:



Two's-Complement Representation

Negative integers use *two's-complement* representation, using the mathematical principle that the two's complement of an integer is its additive inverse. (If you add a number to its additive inverse, the sum is zero.)

Two's-complement representation is useful to processor designers because it removes the need for separate digital circuits to handle both addition and subtraction. For example, if presented with the expression $A - B$, the processor can simply convert it to an addition expression: $A + (-B)$.

The two's complement of a binary integer is formed by inverting (complementing) its bits and adding 1. Using the 8-bit binary value 00000001, for example, its two's complement turns out to be 11111111, as can be seen as follows:

Starting value	00000001
Step 1: Reverse the bits	11111110
Step 2: Add 1 to the value from Step 1	11111110 +00000001
Sum: Two's-complement representation	11111111

11111111 is the two's-complement representation of −1. The two's-complement operation is reversible, so the two's complement of 11111111 is 00000001.

Hexadecimal Two's Complement To create the two's complement of a hexadecimal integer, reverse all bits and add 1. An easy way to reverse the bits of a hexadecimal digit is to subtract the digit from 15. Here are examples of hexadecimal integers converted to their two's complements:

```
6A3D --> 95C2 + 1 --> 95C3
95C3 --> 6A3C + 1 --> 6A3D
```

Converting Signed Binary to Decimal Use the following algorithm to calculate the decimal equivalent of a signed binary integer:

- If the highest bit is a 1, the number is stored in two's-complement notation. Create its two's complement a second time to get its positive equivalent. Then convert this new number to decimal as if it were an unsigned binary integer.
- If the highest bit is a 0, you can convert it to decimal as if it were an unsigned binary integer.

For example, signed binary 11110000 has a 1 in the highest bit, indicating that it is a negative integer. First we create its two's complement, and then convert the result to decimal. Here are the steps in the process:

Starting value	11110000
Step 1: Reverse the bits	00001111
Step 2: Add 1 to the value from Step 1	00001111 + 1
Step 3: Create the two's complement	00010000
Step 4: Convert to decimal	16

Because the original integer (11110000) was negative, we know that its decimal value is –16.

Converting Signed Decimal to Binary To create the binary representation of a signed decimal integer, do the following:

1. Convert the absolute value of the decimal integer to binary.
2. If the original decimal integer was negative, create the two's complement of the binary number from the previous step.

For example, –43 decimal is translated to binary as follows:

1. The binary representation of unsigned 43 is 00101011.
2. Because the original value was negative, we create the two's complement of 00101011, which is 11010101. This is the representation of –43 decimal.

Converting Signed Decimal to Hexadecimal To convert a signed decimal integer to hexadecimal, do the following:

1. Convert the absolute value of the decimal integer to hexadecimal.
2. If the decimal integer was negative, create the two's complement of the hexadecimal number from the previous step.

Converting Signed Hexadecimal to Decimal To convert a signed hexadecimal integer to decimal, do the following:

1. If the hexadecimal integer is negative, create its two's complement; otherwise, retain the integer as is.
2. Using the integer from the previous step, convert it to decimal. If the original value was negative, attach a minus sign to the beginning of the decimal integer.

> You can tell whether a hexadecimal integer is positive or negative by inspecting its most significant (highest) digit. If the digit is ≥ 8, the number is negative; if the digit is ≤ 7, the number is positive. For example, hexadecimal 8A20 is negative and 7FD9 is positive.

Maximum and Minimum Values

A signed integer of n bits uses only $n - 1$ bits to represent the number's magnitude. Table 1-7 shows the minimum and maximum values for signed bytes, words, doublewords, and quadwords.

Table 1-7 Ranges and Sizes of Signed Integer Types.

Type	Range	Storage Size in Bits
Signed byte	-2^7 to $+2^7 - 1$	8
Signed word	-2^{15} to $+2^{15} - 1$	16
Signed doubleword	-2^{31} to $+2^{31} - 1$	32
Signed quadword	-2^{63} to $+2^{63} - 1$	64
Signed double quadword	-2^{127} to $+2^{127} - 1$	128

1.3.7 Binary Subtraction

Subtracting a smaller unsigned binary number from a large one is easy if you go about it in the same way you handle decimal subtraction. Here's an example:

```
  0 1 1 0 1     (decimal 13)
- 0 0 1 1 1     (decimal 7)
-----------
```

Subtracting the bits in position 0 is straightforward:

```
  0 1 1 0 1
- 0 0 1 1 1
-----------
          0
```

In the next position (0 – 1), we are forced to borrow a 1 from the next position to the left. Here's the result of subtracting 1 from 2:

```
  0 1 0 0 1
- 0 0 1 1 1
-----------
        1 0
```

In the next bit position, we again have to borrow a bit from the column just to the left and subtract 1 from 2:

```
  0 0 0 1 1
- 0 0 1 1 1
-----------
      1 1 0
```

Finally, the two high-order bits are zero minus zero:

```
  0 0 0 1 1
- 0 0 1 1 1
-----------
  0 0 1 1 0      (decimal 6)
```

A simpler way to approach binary subtraction is to reverse the sign of the value being subtracted, and then add the two values. This method requires you to have an extra empty bit to hold the number's sign. Let's try it with the same problem we just calculated: (01101 minus 00111). First, we negate 00111 by inverting its bits (11000) and adding 1, producing 11001. Next, we add the binary values and ignore the carry out of the highest bit:

```
0 1 1 0 1       (+13)
1 1 0 0 1       (-7)
---------
0 0 1 1 0       (+6)
```

The result, +6, is exactly what we expected.

1.3.8 Character Storage

If computers only store binary data, how do they represent characters? They use a *character set*, which is a mapping of characters to integers. In earlier times, character sets used only 8 bits. Even now, when running in character mode (such as MS-DOS), IBM-compatible microcomputers use the ***ASCII*** (pronounced "askey") character set. ASCII is an acronym for *American Standard Code for Information Interchange*. In ASCII, a unique 7-bit integer is assigned to each character. Because ASCII codes use only the lower 7 bits of every byte, the extra bit is used on various computers to create a proprietary character set. On IBM-compatible microcomputers, for example, values 128 through 255 represent graphic symbols and Greek characters.

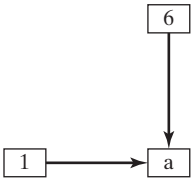
ANSI Character Set The American National Standards Institute (ANSI) defines an 8-bit character set that represents up to 256 characters. The first 128 characters correspond to the letters and symbols on a standard U.S. keyboard. The second 128 characters represent special characters such as letters in international alphabets, accents, currency symbols, and fractions. Early version of Microsoft Windows used the ANSI character set.

Unicode Standard Today, computers must be able to represent a wide variety of international languages in computer software. As a result, the *Unicode* standard was created as a universal way of defining characters and symbols. It defines numeric codes (called *code points*) for characters, symbols, and punctuation used in all major languages, as well as European alphabetic scripts, Middle Eastern right-to-left scripts, and many scripts of Asia. Three transformation formats are used to transform code points into displayable characters:

- **UTF-8** is used in HTML, and has the same byte values as ASCII.
- **UTF-16** is used in environments that balance efficient access to characters with economical use of storage. Recent versions of Microsoft Windows, for example, use UTF-16 encoding. Each character is encoded in 16 bits.
- **UTF-32** is used in environments where space is no concern and fixed-width characters are required. Each character is encoded in 32 bits.

ASCII Strings A sequence of one or more characters is called a *string*. More specifically, an *ASCII string* is stored in memory as a succession of bytes containing ASCII codes. For example, the numeric codes for the string "ABC123" are 41h, 42h, 43h, 31h, 32h, and 33h. A *null-terminated* string is a string of characters followed by a single byte containing zero. The C and C++ languages use null-terminated strings, and many Windows operating system functions require strings to be in this format.

Using the ASCII Table A table on pages 716–717 of this book lists ASCII codes used when running in Windows Console mode. To find the hexadecimal ASCII code of a character, look along the top row of the table and find the column containing the character you want to translate. The most significant digit of the hexadecimal value is in the second row at the top of the table; the least significant digit is in the second column from the left. For example, to find the ASCII code of the letter **a**, find the column containing the **a** and look in the second row: The first hexadecimal digit is 6. Next, look to the left along the row containing **a** and note that the second column contains the digit 1. Therefore, the ASCII code of **a** is 61 hexadecimal. This is shown as follows in simplified form:



ASCII Control Characters Character codes in the range 0 through 31 are called *ASCII control characters.* If a program writes these codes to standard output (as in C++), the control characters will carry out predefined actions. Table 1-8 lists the most commonly used characters in this range, and a complete list may be found on pages 714–715 of this book.

Table 1-8 ASCII Control Characters.

ASCII Code (Decimal)	Description
8	Backspace (moves one column to the left)
9	Horizontal tab (skips forward *n* columns)
10	Line feed (moves to next output line)
12	Form feed (moves to next printer page)
13	Carriage return (moves to leftmost output column)
27	Escape character

Terminology for Numeric Data Representation It is important to use precise terminology when describing the way numbers and characters are represented in memory and on the display screen. Decimal 65, for example, is stored in memory as a single binary byte as 01000001. A debugging program would probably display the byte as "41," which is the number's hexadecimal representation. If the byte were copied to video memory, the letter "**A**" would appear on the screen because 01000001 is the ASCII code for the letter **A**. Because a number's interpretation can depend on the context in which it appears, we assign a specific name to each type of data representation to clarify future discussions:

- A *binary integer* is an integer stored in memory in its raw format, ready to be used in a calculation. Binary integers are stored in multiples of 8 bits (such as 8, 16, 32, or 64).

- A *digit string* is a string of ASCII characters, such as "123" or "65." This is simply a representation of the number and can be in any of the formats shown for the decimal number 65 in Table 1-9:

Table 1-9 Types of Digit Strings.

Format	Value
Binary digit string	"01000001"
Decimal digit string	"65"
Hexadecimal digit string	"41"
Octal digit string	"101"

1.3.9 Section Review

1. Explain the term *least significant bit* (LSB).
2. What is the decimal representation of each of the following unsigned binary integers?
   a. 11111000
   b. 11001010
   c. 11110000
3. What is the sum of each pair of binary integers?
   a. 00001111 + 00000010
   b. 11010101 + 01101011
   c. 00001111 + 00001111
4. How many bytes are contained in each of the following data types?
   a. word
   b. doubleword
   c. quadword
   d. double quadword
5. What is the minimum number of binary bits needed to represent each of the following unsigned decimal integers?
   a. 65
   b. 409
   c. 16385
6. What is the hexadecimal representation of each of the following binary numbers?
   a. 0011 0101 1101 1010
   b. 1100 1110 1010 0011
   c. 1111 1110 1101 1011
7. What is the binary representation of the following hexadecimal numbers?
   a. A4693FBC
   b. B697C7A1
   c. 2B3D9461

1.4 Boolean Expressions

Boolean algebra defines a set of operations on the values **true** and **false**. It was invented by George Boole, a mid-nineteenth-century mathematician. When early digital computers were invented, it was found that Boole's algebra could be used to describe the design of digital circuits. At the same time, boolean expressions are used in computer programs to express logical operations.

A *boolean expression* involves a boolean operator and one or more operands. Each boolean expression implies a value of true or false. The set of operators includes the following:

- NOT: notated as ¬ or ~ or '
- AND: notated as ∧ or •
- OR: notated as ∨ or +

The NOT operator is unary, and the other operators are binary. The operands of a boolean expression can also be boolean expressions. The following are examples:

Expression	Description
¬X	NOT X
X ∧ Y	X AND Y
X ∨ Y	X OR Y
¬X ∨ Y	(NOT X) OR Y
¬(X ∧ Y)	NOT (X AND Y)
X ∧ ¬Y	X AND (NOT Y)

NOT The NOT operation reverses a boolean value. It can be written in mathematical notation as ¬X, where X is a variable (or expression) holding a value of true (T) or false (F). The following truth table shows all the possible outcomes of NOT using a variable **X**. Inputs are on the left side and outputs (shaded) are on the right side:

X	¬X
F	T
T	F

A truth table can use 0 for false and 1 for true.

AND The Boolean AND operation requires two operands, and can be expressed using the notation X ∧ Y. The following truth table shows all the possible outcomes (shaded) for the values of X and Y:

X	Y	X ∧ Y
F	F	F
F	T	F
T	F	F
T	T	T

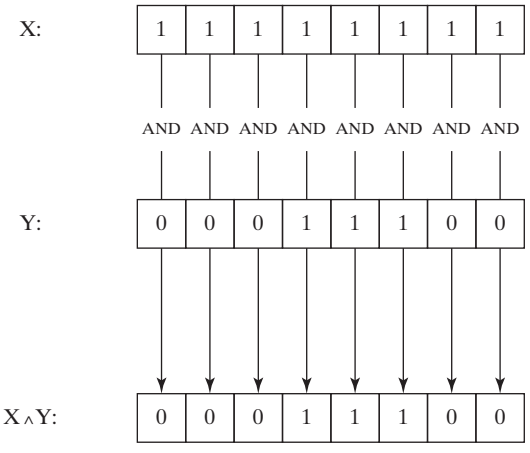
The output is true only when both inputs are true. This corresponds to the logical AND used in compound boolean expressions in C++ and Java.

The AND operation is often carried out at the bit level in assembly language. In the following example, each bit in X is ANDed with its corresponding bit in Y:

```
X:          11111111
Y:          00011100
X ∧ Y:      00011100
```

As Figure 1-2 shows, each bit of the resulting value, 00011100, represents the result of ANDing the corresponding bits in X and Y.

Figure 1–2 ANDing the bits of two binary integers.



OR The Boolean OR operation requires two operands, and is often expressed using the notation **X** ∨ **Y**. The following truth table shows all the possible outcomes (shaded) for the values of X and Y:

X	**Y**	**X ∨ Y**
F	F	F
F	T	T
T	F	T
T	T	T

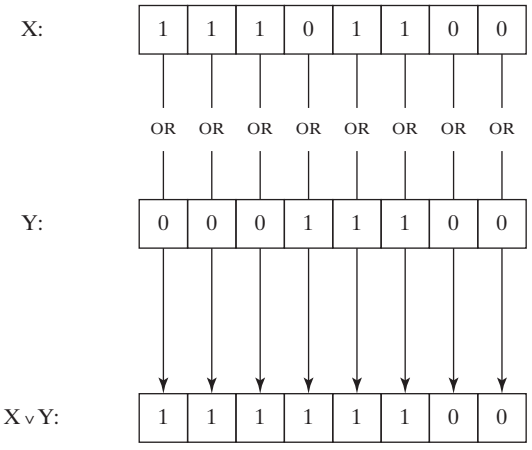
The output is false only when both inputs are false. This truth table corresponds to the logical OR used in compound boolean expressions in C++ and Java.

The OR operation is often carried out at the bit level. In the following example, each bit in X is ORed with its corresponding bit in Y, producing 11111100:

```
X:          11101100
Y:          00011100
X ∨ Y:      11111100
```

As shown in Figure 1-3, the bits are ORed individually, producing a corresponding bit in the result.

Figure 1–3 ORing the bits in two binary integers.



Operator Precedence *Operator precedence rules* are used to indicate which operators execute first in expressions involving multiple operators. In a boolean expression involving more than one operator, precedence is important. As shown in the following table, the NOT operator has the highest precedence, followed by AND and OR. You can use parentheses to force the initial evaluation of an expression:

Expression	Order of Operations
¬X ∨ Y	NOT, then OR
¬(X ∨ Y)	OR, then NOT
X ∨ (Y ∧ Z)	AND, then OR

1.4.1 Truth Tables for Boolean Functions

A *boolean function* receives boolean inputs and produces a boolean output. A truth table can be constructed for any boolean function, showing all possible inputs and outputs. The following are truth tables representing boolean functions having two inputs named X and Y. The shaded column on the right is the function's output:

Example 1: $\neg X \vee Y$

X	$\neg X$	Y	$\neg X \vee Y$
F	T	F	T
F	T	T	T
T	F	F	F
T	F	T	T

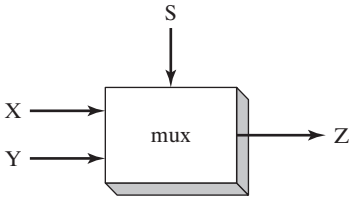
Example 2: $X \wedge \neg Y$

X	Y	$\neg Y$	$X \wedge \neg Y$
F	F	T	F
F	T	F	F
T	F	T	T
T	T	F	F

Example 3: $(Y \wedge S) \vee (X \wedge \neg S)$

X	Y	S	$Y \wedge S$	$\neg S$	$X \wedge \neg S$	$(Y \wedge S) \vee (X \wedge \neg S)$
F	F	F	F	T	F	F
F	T	F	F	T	F	F
T	F	F	F	T	T	T
T	T	F	F	T	T	T
F	F	T	F	F	F	F
F	T	T	T	F	F	T
T	F	T	F	F	F	F
T	T	T	T	F	F	T

The boolean function in Example 3 describes a *multiplexer*, a digital component that uses a selector bit (S) to select one of two outputs (X or Y). If S = false, the function output (Z) is the same as X. If S = true, the function output is the same as Y. Here is a block diagram of a multiplexer:



1.4.2 Section Review

1. Describe the following boolean expression: $\neg X \vee Y$.
2. Describe the following boolean expression: $(X \wedge Y)$.
3. What is the value of the boolean expression $(T \wedge F) \vee T$?
4. What is the value of the boolean expression $\neg(F \vee T)$?
5. What is the value of the boolean expression $\neg F \vee \neg T$?

1.5 Chapter Summary

This book focuses on programming x86 processors, using the MS-Windows platform. We cover basic principles about computer architecture, machine language, and low-level programming. You will learn enough assembly language to test your knowledge on today's most widely used microprocessor family.

Before reading this book, you should have completed a single college course or equivalent in computer programming.

An *assembler* is a program that converts source-code programs from assembly language into machine language. A companion program, called a linker, combines individual files created by an assembler into a single executable program. A third program, called a debugger, provides a way for a programmer to trace the execution of a program and examine the contents of memory.

You will create 32-bit and 64-bit programs for the most part, and 16-bit programs if you focus on the last four chapters.

You will learn the following concepts from this book: basic computer architecture applied to x86 (and Intel 64) processors; elementary boolean logic; how x86 processors manage memory; how high-level language compilers translate statements from their language into assembly language and native machine code; how high-level languages implement arithmetic expressions, loops, and logical structures at the machine level; and the data representation of signed and unsigned integers, real numbers, and character data.

Assembly language has a *one-to-one* relationship with machine language, in which a single assembly language instruction corresponds to one machine language instruction. Assembly language is not portable because it is tied to a specific processor family.

Programming languages are tools that you can use to create individual applications or parts of applications. Some applications, such as device drivers and hardware interface routines, are more suited to assembly language. Other applications, such as multiplatform commercial and scientific applications, are more easily written in high-level languages.

The *virtual machine* concept is an effective way of showing how each layer in a computer architecture represents an abstraction of a machine. Layers can be constructed of hardware or software, and programs written at any layer can be translated or interpreted by the next-lowest layer. The virtual machine concept can be related to real-world computer layers, including digital logic, instruction set architecture, assembly language, and high-level languages.

Binary and hexadecimal numbers are essential notational tools for programmers working at the machine level. For this reason, you must understand how to manipulate and translate between number systems and how character representations are created by computers.

The following boolean operators were presented in this chapter: NOT, AND, and OR. A boolean expression combines a boolean operator with one or more operands. A truth table is an effective way to show all possible inputs and outputs of a boolean function.

1.6 Key Terms

ASCII
ASCII control characters
ASCII digit string
assembler
assembly language
binary digit string
binary integer
bit
boolean algebra
boolean expression
boolean function
character set
code interpretation
code point (Unicode)
code translation
debugger
device driver
digit string
embedded systems application
exabyte
gigabyte
hexadecimal digit string
hexadecimal integer
high-level language
instruction set architecture (ISA)
Java Native Interface (JNI)
kilobyte
language portability
least significant bit (LSB)
machine language
megabyte
microcode interpreter
microprogram
Microsoft Macro Assembler (MASM)
most significant bit (MSB)
multiplexer
null-terminated string
octal digit string
one-to-many relationship
operator precedence
petabyte
registers
signed binary integer
terabyte
Unicode
Unicode Transformation Format (UTF)

unsigned binary integer
UTF-8
UTF-16
UTF-32
virtual machine (VM)
virtual machine concept
Visual Studio
yottabyte
zettabyte

1.7 Review Questions and Exercises

1.7.1 Short Answer

1. In an 8-bit binary number, which is the most significant bit (MSB)?
2. What are the hexadecimal and decimal representations of each of the following unsigned binary integers?

   a. `1000 1010 0011 0001`
   b. `1010 1110 0000 1010`
   c. `1110 1010 1111 0011`

3. What is the sum of each pair of binary integers?

   a. `10101111 + 11011011`
   b. `10010111 + 11111111`
   c. `01110101 + 10101100`

4. Calculate unsigned hexadecimal 03h plus 2345h.
5. How many bits are used by each of the following data types?
   a. word
   b. doubleword
   c. quadword
   d. double quadword
6. What is the binary representation of each of the following decimal numbers?
   a. 123
   b. 230
   c. 190
7. What is the hexadecimal representation of each of the following binary numbers?
   a. 0011 0101 1101 1010
   b. 1100 1110 1010 0011
   c. 1111 1110 1101 1011
8. What is the binary representation of the following hexadecimal numbers?
   a. 0126F9D4
   b. 6ACDFA95
   c. F69BDC2A
9. What is the unsigned decimal representation of each of the following hexadecimal integers?
   a. 3A
   b. 1BF
   c. 1001

10. What is the unsigned decimal representation of each of the following hexadecimal integers?
    a. 1234
    b. 2323
    c. 408
11. What is the 16-bit hexadecimal representation of each of the following signed decimal integers?
    a. –24
    b. –331
12. What is the 16-bit hexadecimal representation of each of the following signed decimal integers?
    a. –42
    b. –1276
13. The following 16-bit hexadecimal numbers represent signed integers. Convert each to decimal.
    a. 6BF9
    b. C123
14. The following 16-bit hexadecimal numbers represent signed integers. Convert each to decimal.
    a. 4CD2
    b. 8230
15. What is the sum of the binary representations of each of the following pairs of decimal numbers? Verify, in each case, that your answer is identical to the binary representation of their sum.
    a. 15 and 35
    b. 78 and 678
    c. 345 and 567
16. What is the decimal representation of each of the following signed binary numbers?
    a. 10000000
    b. 11001100
    c. 10110111
17. What is the 8-bit binary (two's-complement) representation of each of the following signed decimal integers?
    a. –5
    b. –42
    c. –16
18. What is the 8-bit binary (two's-complement) representation of each of the following signed decimal integers?
    a. –72
    b. –98
    c. –26
19. The following hexadecimal (two's-complement) numbers represent signed integers. Convert each to decimal.
    a. FEE2h
    b. F3h

20. What is the sum of each pair of hexadecimal integers?
    a. 7C4 + 3BE
    b. B69 + 7AD
21. What are the hexadecimal and decimal representations of the ASCII characters & and $?
22. What are the hexadecimal and decimal representations of the ASCII characters m, F, and G?
23. *Challenge:* What is the largest decimal value you can represent, using a 129-bit unsigned integer?
24. Convert 96 and 45 to packed BCD representation. Add the resultant BCD numbers, and then convert back to decimal.
25. Using a truth table that shows all possible inputs and outputs for the boolean functions described by $\neg(A \vee B)$ and $(\neg A \wedge \neg B)$, compare the functions. Have you heard of a mathematician named *Augustus De Morgan?*
26. Make truth tables for the following logic gates for the case of three inputs and one output.
    a. AND
    b. XOR
    c. NAND
27. If a boolean function has four inputs, how many rows are required for its truth table?
28. How many selector bits are required for a four-input multiplexer?

1.7.2 Algorithm Workbench

Use any high-level programming language you wish for the following programming exercises. Do not call built-in library functions that accomplish these tasks automatically. (Examples are sprintf and sscanf from the Standard C library.)

1. Write a function that receives a 16-bit binary integer. The function must return the decimal representation of its square.
2. Write a function that receives a string containing a 32-bit hexadecimal integer. The function must return the string's integer value.
3. Write a function that receives an integer. The function must return a string containing the binary representation of the integer.
4. Write a function that receives a string containing a number represented using base 8. The function must return its decimal equivalent.
5. Write a function that adds two digit strings in base b, where $2 \le b \le 10$. Each string may contain as many as 1,000 digits. Return the sum in a string that uses the same number base.
6. Write a function that adds two hexadecimal strings, each as long as 1,000 digits. Return a hexadecimal string that represents the sum of the inputs.
7. Write a function that multiplies a single hexadecimal digit by a hexadecimal digit string as long as 1,000 digits. Return a hexadecimal string that represents the product.

8. Write a Java program that contains the calculation shown below. Then, use the *javap –c* command to disassemble your code. Add comments to each line that provide your best guess as to its purpose.

```
int Y;
int X = (Y + 4) * 3;
```

9. Devise a way of subtracting unsigned binary integers. Test your technique by subtracting binary 00000101 from binary 10001000, producing 10000011. Test your technique with at least two other sets of integers, in which a smaller value is always subtracted from a larger one.

Chapter End Notes

1. Donald Knuth, MMIX, *A RISC Computer for the New Millennium*, Transcript of a lecture given at the Massachusetts Institute of Technology, December 30, 1999.

2

x86 Processor Architecture

This chapter focuses on the underlying hardware associated with x86 assembly language. It may be said that assembly language is the ideal software tool for communicating directly with a machine. If that is true, then assembly programmers must be intimately familiar with the processor's internal architecture and capabilities. We will discuss some of the basic operations that take place inside the processor when instructions are executed. We will talk about how programs are loaded and executed by the operating system. A sample motherboard layout will give some insight into the hardware environment of x86 systems, and the chapter ends with a discussion of how layered input/output works between application programs and operating systems. All of the topics in this chapter provide the hardware foundation for you to begin writing assembly language programs.

2.1 General Concepts

This chapter describes the architecture of the x86 processor family and its host computer system from a programmer's point of view. Included in this group are all Intel IA-32 and Intel 64 processors, such as the Intel Pentium and Core-Duo, as well as the Advanced Micro Devices (AMD) processors, such as Athlon, Phenom, Opteron, and AMD64. Assembly language is a great tool for learning how a computer works, and it requires you to have a working knowledge of computer hardware. To that end, the concepts and details in this chapter will help you to understand the assembly language code you write.

We strike a balance between concepts applying to all microcomputer systems and specifics about x86 processors. You may work on various processors in the future, so we expose you to broad concepts. To avoid giving you a superficial understanding of machine architecture, we focus on specifics of the x86, which will give you a solid grounding when programming in assembly language.

> If you want to learn more about the Intel IA-32 architecture, read *the Intel 64 and IA-32 Architectures Software Developer's Manual, Volume 1: Basic Architecture.* It's a free download from the Intel web site (www.intel.com).

2.1.1 Basic Microcomputer Design

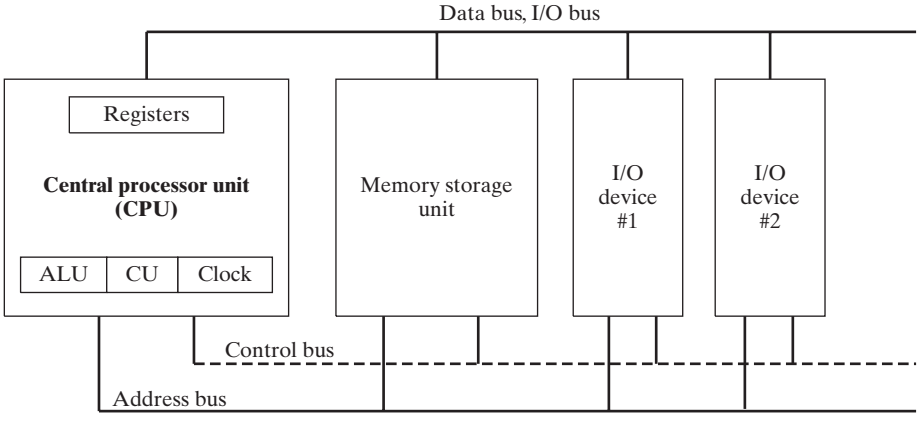
Figure 2-1 shows the basic design of a hypothetical microcomputer. The *central processor unit* (CPU), where calculations and logical operations take place, contains a limited number of storage locations named *registers*, a high-frequency clock, a control unit, and an arithmetic logic unit.

- The *clock* synchronizes the internal operations of the CPU with other system components.
- The *control unit* (CU) coordinates the sequencing of steps involved in executing machine instructions.
- The *arithmetic logic unit* (ALU) performs arithmetic operations such as addition and subtraction and logical operations such as AND, OR, and NOT.

The CPU is attached to the rest of the computer via pins attached to the CPU socket in the computer's motherboard. Most pins connect to the data bus, the control bus, and the address bus. The *memory storage unit* is where instructions and data are held while a computer program is running. The storage unit receives requests for data from the CPU, transfers data from random access memory (RAM) to the CPU, and transfers data from the CPU into memory. All processing of data takes place within the CPU, so programs residing in memory must be copied into the CPU before they can execute. Individual program instructions can be copied into the CPU one at a time, or groups of instructions can be copied together.

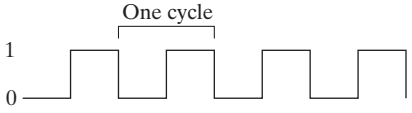
A *bus* is a group of parallel wires that transfer data from one part of the computer to another. A computer system usually contains four bus types: data, I/O, control, and address. The *data bus* transfers instructions and data between the CPU and memory. The I/O bus transfers data

FIGURE 2–1 Block diagram of a microcomputer.



between the CPU and the system input/output devices. The *control bus* uses binary signals to synchronize actions of all devices attached to the system bus. The *address bus* holds the addresses of instructions and data when the currently executing instruction transfers data between the CPU and memory.

Clock Each operation involving the CPU and the system bus is synchronized by an internal clock pulsing at a constant rate. The basic unit of time for machine instructions is a *machine cycle* (or *clock cycle*). The length of a clock cycle is the time required for one complete clock pulse. In the following figure, a clock cycle is depicted as the time between one falling edge and the next:



The duration of a clock cycle is calculated as the reciprocal of the clock's speed, which in turn is measured in oscillations per second. A clock that oscillates 1 billion times per second (1 GHz), for example, produces a clock cycle with a duration of one billionth of a second (1 nanosecond).

A machine instruction requires at least one clock cycle to execute, and a few require in excess of 50 clocks (the multiply instruction on the 8088 processor, for example). Instructions requiring memory access often have empty clock cycles called *wait states* because of the differences in the speeds of the CPU, the system bus, and memory circuits.

2.1.2 Instruction Execution Cycle

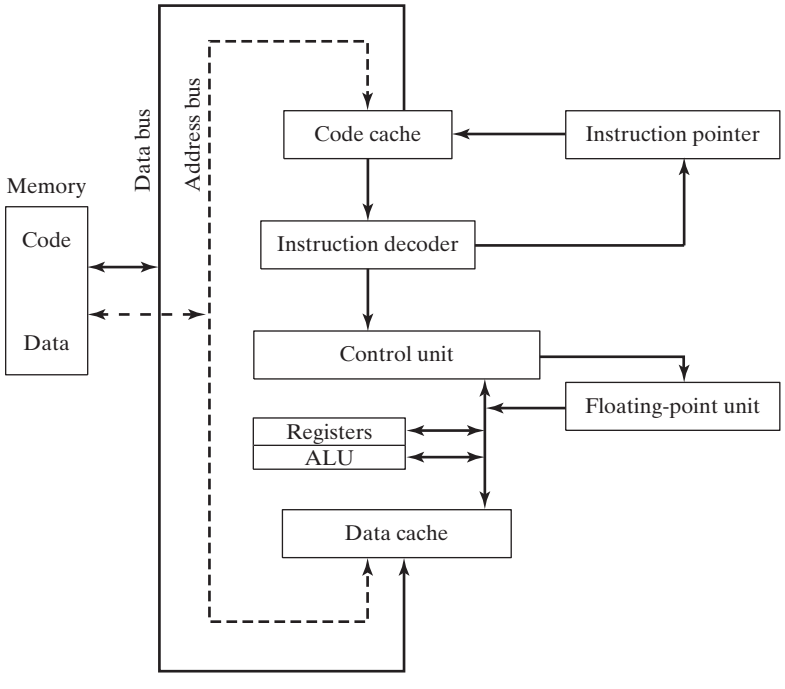
A single machine instruction does not just magically execute all at once. The CPU has to go through a predefined sequence of steps to execute a machine instruction, called the *instruction execution cycle*. Let's assume that the instruction pointer register holds the address of the instruction we want to execute. Here are the steps to execute it:

1. First, the CPU has to **fetch the instruction** from an area of memory called the *instruction queue*. Right after doing this, it increments the instruction pointer.
2. Next, the CPU **decodes** the instruction by looking at its binary bit pattern. This bit pattern might reveal that the instruction has operands (input values).
3. If operands are involved, the CPU **fetches the operands** from registers and memory. Sometimes, this involves address calculations.
4. Next, the CPU **executes** the instruction, using any operand values it fetched during the earlier step. It also updates a few status flags, such as Zero, Carry, and Overflow.
5. Finally, if an output operand was part of the instruction, the CPU **stores the result** of its execution in the operand.

We usually simplify this complicated-sounding process to three basic steps: **Fetch**, **Decode**, and **Execute**. An *operand* is a value that is either an input or an output to an operation. For example, the expression Z = X + Y has two input operands (X and Y) and a single output operand (Z).

A block diagram showing data flow within a typical CPU is shown in Figure 2-2. The diagram helps to show relationships between components that interact during the instruction execution cycle. In order to read program instructions from memory, an address is placed on the address bus. Next, the memory controller places the requested code on the data bus, making the code available inside the code cache. The instruction pointer's value determines which instruction will be executed next. The instruction is analyzed by the instruction decoder, causing the appropriate

Figure 2–2 Simplified CPU block diagram.



digital signals to be sent to the control unit, which coordinates the ALU and floating-point unit. Although the control bus is not shown in this figure, it carries signals that use the system clock to coordinate the transfer of data between the different CPU components.

2.1.3 Reading from Memory

As a rule, computers read memory much more slowly than they access internal registers. This is because reading a single value from memory involves four separate steps:

1. Place the address of the value you want to read on the address bus.
2. Assert (change the value of) the processor's RD *(read)* pin.
3. Wait one clock cycle for the memory chips to respond.
4. Copy the data from the data bus into the destination operand.

Each of these steps generally requires a single *clock cycle*, a measurement of time based on a clock that ticks inside the processor at a regular rate. Computer CPUs are often described in terms of their clock speeds. A speed of *1.2 GHz*, for example, means the clock ticks, or oscillates, 1.2 billion times per second. So, 4 clock cycles go by fairly fast, considering each one lasts for only 1/1,200,000,000th of a second. Still, that's much slower than the CPU registers, which are usually accessed in only one clock cycle.

Fortunately, CPU designers figured out a long time ago that computer memory creates a speed bottleneck because most programs have to access variables. They came up with a clever way to reduce the amount of time spent reading and writing memory—they store the most recently used instructions and data in high-speed memory called *cache*. The idea is that a program is more likely to want to access the same memory and instructions repeatedly, so cache keeps these values where they can be accessed quickly. Also, when the CPU begins to execute a program, it can look ahead and load the next thousand instructions (for example) into cache, on the assumption that these instructions will be needed fairly soon. If there happens to be a loop in that block of code, the same instructions will be in cache. When the processor is able to find its data in cache memory, we call that a *cache hit*. On the other hand, if the CPU tries to find something in cache and it's not there, we call that a *cache miss*.

Cache memory for the x86 family comes in two types. *Level-1 cache* (or *primary cache*) is stored right on the CPU. *Level-2 cache* (or *secondary cache*) is a little bit slower, and attached to the CPU by a high-speed data bus. The two types of cache work together in an optimal way.

There's a reason why cache memory is faster than conventional RAM—it's because cache memory is constructed from a special type of memory chip called *static RAM*. It's expensive, but it does not have to be constantly refreshed in order to keep its contents. On the other hand, conventional memory, known as *dynamic RAM*, must be refreshed constantly. It's much slower, but cheaper.

2.1.4 Loading and Executing a Program

Before a program can run, it must be loaded into memory by a utility known as a *program loader*. After loading, the operating system must point the CPU to the program's *entry point*, which is the address at which the program is to begin execution. The following steps break this process down in more detail:

- The operating system (OS) searches for the program's filename in the current disk directory. If it cannot find the name there, it searches a predetermined list of directories (called *paths*) for the filename. If the OS fails to find the program filename, it issues an error message.
- If the program file is found, the OS retrieves basic information about the program's file from the disk directory, including the file size and its physical location on the disk drive.
- The OS determines the next available location in memory and loads the program file into memory. It allocates a block of memory to the program and enters information about the program's size and location into a table (sometimes called a *descriptor table*). Additionally, the OS may adjust the values of pointers within the program so they contain addresses of program data.
- The OS begins execution of the program's first machine instruction (its entry point). As soon as the program begins running, it is called a *process*. The OS assigns the process an identification number (*process ID*), which is used to keep track of it while running.
- The *process* runs by itself. It is the OS's job to track the execution of the process and to respond to requests for system resources. Examples of resources are memory, disk files, and input-output devices.
- When the process ends, it is removed from memory.

> ***Tip:*** If you're using any version of Microsoft Windows, press *Ctrl-Alt-Delete* and select the *Task Manager* item. The Task Manager window lets you view lists of Applications and Processes. Applications are the names of complete programs currently running, such as Windows Explorer or Microsoft Visual C++. When you click on the *Processes* tab, you see a long list of process names. Each of those processes is a small program running independently of all the others. You can continuously track the amount of CPU time and memory used by each process. In some cases, you can shut down a process by selecting its name and pressing the *Delete* key.

2.1.5 Section Review

1. The central processor unit (CPU) contains registers and what other basic elements?
2. The central processor unit is connected to the rest of the computer system using what three buses?
3. Why does memory access take more machine cycles than register access?
4. What are the three basic steps in the instruction execution cycle?
5. Which two additional steps are required in the instruction execution cycle when a memory operand is used?

2.2 32-Bit x86 Processors

In this section, we focus on the basic architectural features of all x86 processors. This includes members of the Intel IA-32 family as well as all 32-bit AMD processors.

2.2.1 Modes of Operation

x86 processors have three primary modes of operation: protected mode, real-address mode, and system management mode. A sub-mode, named *virtual-8086*, is a special case of protected mode. Here are short descriptions of each:

Protected Mode Protected mode is the native state of the processor, in which all instructions and features are available. Programs are given separate memory areas named *segments*, and the processor prevents programs from referencing memory outside their assigned segments.

Virtual-8086 Mode While in protected mode, the processor can directly execute real-address mode software such as MS-DOS programs in a safe environment. In other words, if a program crashes or attempts to write data into the system memory area, it will not affect other programs running at the same time. A modern operating system can execute multiple separate virtual-8086 sessions at the same time.

Real-Address Mode Real-address mode implements the programming environment of an early Intel processor with a few extra features, such as the ability to switch into other modes. This mode is useful if a program requires direct access to system memory and hardware devices.

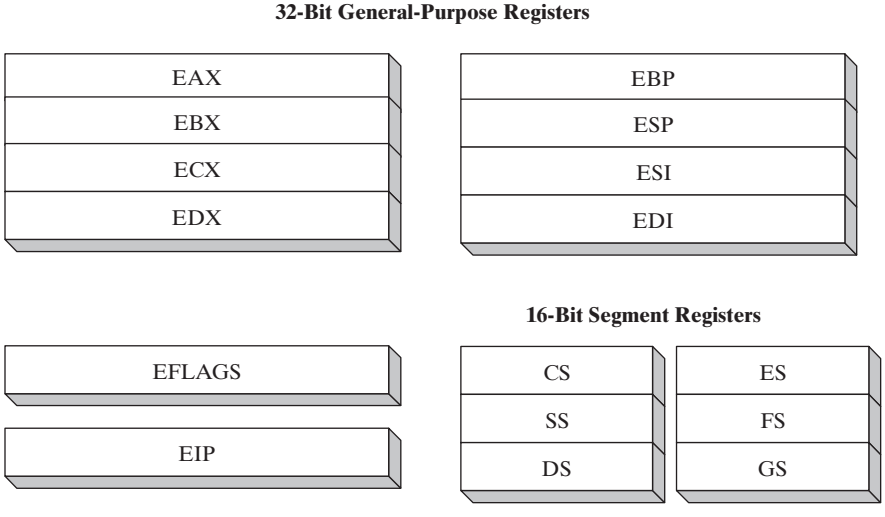
System Management Mode System management mode (SMM) provides an operating system with a mechanism for implementing functions such as power management and system security. These functions are usually implemented by computer manufacturers who customize the processor for a particular system setup.

2.2.2 Basic Execution Environment

Address Space

In 32-bit protected mode, a task or program can address a linear address space of up to 4 GBytes. Beginning with the P6 processor, a technique called *extended physical addressing* allows a total of 64 GBytes of physical memory to be addressed. Real-address mode programs, on the other hand, can only address a range of 1 MByte. If the processor is in protected mode and running multiple programs in virtual-8086 mode, each program has its own 1-MByte memory area.

Figure 2–3 Basic program execution registers.

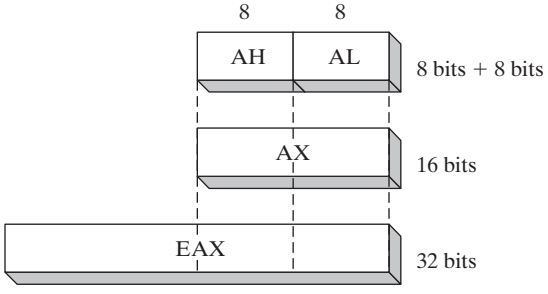


Basic Program Execution Registers

Registers are high-speed storage locations directly inside the CPU, designed to be accessed at much higher speed than conventional memory. When a processing loop is optimized for speed, for example, loop counters are held in registers rather than variables. Figure 2-3 shows the *basic program execution registers*. There are eight general-purpose registers, six segment registers, a processor status flags register (EFLAGS), and an instruction pointer (EIP).

General-Purpose Registers The *general-purpose registers* are primarily used for arithmetic and data movement. As shown in Figure 2-4, the lower 16 bits of the EAX register can be referenced by the name AX.

Figure 2–4 General-purpose registers.



Portions of some registers can be addressed as 8-bit values. For example, the AX register has an 8-bit upper half named AH and an 8-bit lower half named AL. The same overlapping relationship exists for the EAX, EBX, ECX, and EDX registers:

32-Bit	16-Bit	8-Bit (High)	8-Bit (Low)
EAX	AX	AH	AL
EBX	BX	BH	BL
ECX	CX	CH	CL
EDX	DX	DH	DL

The remaining general-purpose registers can only be accessed using 32-bit or 16-bit names, as shown in the following table:

32-Bit	16-Bit
ESI	SI
EDI	DI
EBP	BP
ESP	SP